Xuan Du Trinh

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Google Scholar

Summary

Ph.D. researcher in quantum information theory at Stony Brook University, with prior training in theoretical physics at ENS Paris and in quantum information at Sorbonne Université. My research covers quantum entanglement, quantum networks, open quantum systems, quantum circuit verification and equivalence checking, and quantum channel learning. I build Python/C++ research software and benchmark pipelines, use linear and semidefinite programming in numerical work, and have run circuit experiments on IBM Quantum devices.

Research interests: Quantum networking, quantum information theory, quantum computing, quantum learning theory, quantum programming languages

Education

09/2023 – Present — **Ph.D. in Computer Science**
Stony Brook University, New York, United States
Advisors: *Nengkun Yu, Aruna Balasubramanian*

09/2022 – 08/2023 — **Master of Science in Quantum Information**
Quantum Information Center (QICS), Sorbonne Université, Paris, France
Thesis: "Nonlocality Distillation"
Advisors: *Jean-Daniel Bancal, Mirjam Weilenmann, Peter Brown*

09/2021 – 08/2022 — **Master of Science in Theoretical Physics**
International Centre for Fundamental Physics (ICFP), École normale supérieure (ENS) de Paris, France
Thesis: "Classical Simulation of Shallow Random Quantum Circuits"
Advisor: *Omar Fawzi*

09/2018 – 08/2021 — **Engineering Degree (Ingénieur polytechnicien)**
École Polytechnique, Institut Polytechnique de Paris, Palaiseau, France
Specialization: Fundamental Physics, third-year program "From Particles to Stars"

09/2015 – 06/2018 — **Bachelor of Science, Gifted Program in Physics**
Hanoi University of Science, Hanoi, Vietnam

09/2012 – 05/2015 — **HUS High School for Gifted Students, Mathematics**
Hanoi University of Science, Hanoi, Vietnam

Research Experience

09/2023 – Present — **Research Assistant**
Department of Computer Science, Stony Brook University
Advisors: *Nengkun Yu, Aruna Balasubramanian*

Quantum circuit verification

- Built a Python/C++ tool for scalable equivalence checking and verification of shallow quantum circuits, combining symbolic methods, local-projection fingerprints, and weighted binary decision diagrams (*arXiv:2601.22372*).

Quantum learning and noise

- Study quantum channel learning and Pauli-noise characterization, including sample-complexity analysis and validation of estimators against simulation.
- Run and analyze circuit experiments on IBM Quantum cloud devices through group projects, comparing hardware data with simulator expectations and circuit-level noise models.

Quantum networks

- Certify entanglement between network nodes from measured CHSH violation (QCE 2026).
- Contributed to the design and analysis of a verification-based quantum transmission protocol (*ICDCS 2026*, Distinguished Paper Award).
- Study how entanglement evolves under noise and during entanglement swapping, including limitations of generalized entanglement swapping under noise (*arXiv:2609.11021*).

05/2023 – 09/2023 **Research Internship**
Center for Theoretical Physics (IPhT), Commissariat à l'Énergie Atomique (CEA) Saclay, France
Advisors: *Jean-Daniel Bancal, Mirjam Weilenmann, Peter Brown*
Project: "Nonlocality Distillation"

- Formulated quantum-information constraints as linear programs and developed optimization-based methods for nonlocality distillation, with applications to quantum communication and quantum key distribution. Wrote Python code for numerical optimization experiments.

04/2022 – 07/2022 **Research Internship**
Centre Inria de Lyon, Laboratoire de l'Informatique du Parallélisme (LIP), ENS de Lyon, France
Advisor: *Omar Fawzi*
Project: "Classical Simulation of Shallow Random Quantum Circuits"

- Studied existing techniques for classical simulation of shallow random quantum circuits.
- Developed and tested semidefinite-programming-based simulation methods.

03/2021 – 07/2021 **Research Internship**
Irène Joliot-Curie Laboratory of Physics of Two Infinities (IJCLab), Université Paris-Saclay, Orsay, France
Advisors: *Louis Fayard, Zhiqing Zhang*
Project: "PDF Uncertainties and the W Boson Mass Measurement"

- Analyzed how parton distribution function uncertainties contribute to the systematic uncertainty of the W boson mass measurement in the ATLAS detectors.

Publications

2026 **Emergence of Classical Stochasticity**, ***Xuan Du Trinh***, *Ismaël Septembre, Hai-Chau Nguyen*, *Physical Review A*, 114, 042207. Preprint: *arXiv:2602.04633 [quant-ph]*

2026 **Full-Rank Noise Forbids Long-Range Entanglement Swapping**, ***Xuan Du Trinh***, *Nicholas Pardave, Angie Huang, Xiangyi Meng, Nengkun Yu*, *arXiv:2609.11021 [quant-ph]*

2026 **One Relative Phase Orders Operational Thresholds of Noisy Bell Pairs**, ***Xuan Du Trinh***, *arXiv:2609.10034 [quant-ph]*

2026 **Operational Thresholds of Bell Mixtures with Complex X Noise**, ***Xuan Du Trinh***, *arXiv:2608.17609 [quant-ph]*

2026 **Absorption Capacity of Separable Noise: Bell-Mixing Thresholds on Entanglement and Teleportation**, ***Xuan Du Trinh***, *arXiv:2606.31243 [quant-ph]*

2026 **Entanglement Certification by Measuring Nonlocality**, ***Xuan Du Trinh***, *Zhengyu Wu, Junlin Bai, Huan-Hsin Tseng, Nengkun Yu, Aruna Balasubramanian*, *IEEE International Conference on Quantum Computing and Engineering (IEEE Quantum Week, QCE 2026)*, QNET track. *arXiv:2507.18066 [quant-ph]*

2026 **Rethinking Quantum Network Design Using a Verification-Based Quantum Transmission Protocol**, *Yiming Zeng, Zhengyu Wu,* ***Xuan Du Trinh****, Yuanyuan Yang, Nengkun Yu, Aruna Balasubramanian*, *46th IEEE International Conference on Distributed Computing Systems (ICDCS 2026)*. **Distinguished Paper Award**

2026 **Manjushri: A Tool for Equivalence Checking of Quantum Circuits**, ***Xuan Du Trinh***, *Meghana Sistla, Nengkun Yu, Thomas Reps*, *arXiv:2601.22372 [quant-ph]*

2025 **Scalable Equivalence Checking and Verification of Shallow Quantum Circuits**, *Nengkun Yu,* ***Xuan Du Trinh****, Thomas Reps*, *Proceedings of the ACM on Programming Languages*, Vol. 9, OOPSLA2

2025 **Adaptivity Is Not Helpful for Pauli Channel Learning**, ***Xuan Du Trinh***, *Nengkun Yu*, *Quantum*, 9, 1864

Talks

09/2026 **Entanglement Certification by Measuring Nonlocality**
IEEE Quantum Week (QCE 2026), QNET track, Toronto, Canada

10/2025 **Scalable Equivalence Checking and Verification of Shallow Quantum Circuits**
ACM SPLASH / OOPSLA 2025, Singapore

Teaching and Mentoring

Fall 2025 **Teaching Assistant, Analysis of Algorithms (graduate)**
Stony Brook University. Algorithms and data structures. Prepared homework and exam materials, held weekly office hours, graded assignments, and helped students with problem solving, debugging, and exam preparation.

Summer 2025 **Research Mentor, High School Student**
Mentored a high school student through a multi-month research project, with weekly guidance on research planning, writing, and technical communication. The student was selected as a **Top 300 Scholar** in the **85th Regeneron Science Talent Search** (2,612 entrants), was admitted to MIT, Harvard, and Yale, and will attend Harvard.

Fall 2023 **Teaching Assistant, Foundations of Computer Science (undergraduate)**
Stony Brook University. Discrete mathematics and proof techniques. Prepared homework and exam materials, held office hours, graded assignments, and helped students with problem solving and exam preparation.

Service

06/2024 **Reviewer**, *44th IEEE International Conference on Distributed Computing Systems (ICDCS 2024)*

02/2024 **Reviewer**, *International Joint Conference on Neural Networks (IJCNN 2024), Quantum Machine Learning Track*

Honors and Awards

2026 **Distinguished Paper Award**
46th IEEE International Conference on Distributed Computing Systems (ICDCS 2026)

09/2023 – 09/2025 **Chairman's Fellowship**
Department of Computer Science, Stony Brook University

10/2021 – 09/2022 **VALLET Scholarship**
VALLET Foundation, France

09/2018 – 08/2021 **EIFFEL Scholarship**
French Government, for outstanding international students

Skills

Programming	Python, C++, C, Java, Julia, MATLAB, Bash, SQL
Libraries	NumPy, SciPy, Pandas, Matplotlib, Qiskit, Quasimodo
Data	Data processing and cleaning, statistical analysis, benchmarking pipelines, SQL (IBM Db2)
Scientific software	Mathematica, ROOT, COMSOL Multiphysics, Ansys Fluent
Graduate coursework	Quantum information theory, quantum cryptography, quantum algorithms, quantum optics, photonic quantum computing, atomic physics for quantum information, statistical physics, statistical field theory, quantum field theory, Yang–Mills theory, general relativity, quantum metrology, Lie groups, Lie algebras and representations
Other coursework	Natural language processing (neural networks, data preprocessing, model evaluation), database systems, statistics, programming languages, system security
Languages	Vietnamese (Native), English (Proficient), French (Proficient)